

Abhinav Mahadik

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SUMMARY

AI Software Engineer who builds LLM-powered agents and RAG systems that replace manual data workflows. Shipped production security middleware, UI, and data catalog systems for a B2B Food & Business platform. MS in Computer Science with a Graduate Certificate in Deep Learning from the University of Texas, USA.

TECHNICAL SKILLS

GenAI & LLMs: OpenAI APIs, RAG, Ollama, Agentic AI, Multi-Agent Systems, Prompt Engineering, PEFT/LoRA
AI/ML Frameworks: LangChain, LangGraph, HuggingFace Transformers, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas
Languages: Python (Proficient), SQL, JavaScript, TypeScript, C++
Web & Backend: FastAPI, Flask, NestJS, Angular, React, Next.js, Vue.js, Node.js, Streamlit
Data & Infrastructure: Pinecone, PostgreSQL, MongoDB, Redis, SQLite, Docker, Git, GitHub Actions (CI/CD)

EXPERIENCE

Software Engineer

Nov 2025 – Present

The Feast

Mumbai, MH

- Built a catalog export feature across 8+ relational tables with GraphQL filters on Hasura and a Vue.js frontend for localized Excel generation, cutting reporting turnaround from 4 hours to under 30 minutes.
- Shipped a multi-step auth system in NestJS – chose OTP-over-email with AWS SES verification to reduce cost per auth event, added JWT refresh rotation and Redis-backed rate limiting for brute-force protection.
- Designed a security middleware layer using Angular HTTP interceptors and role-aware guards that validate session state on every API call, closing an access-control gap found during internal security review.
- Developed and optimized MongoDB aggregation pipelines for cross-branch inventory and order data, implementing indexed lookups, schema-level validation, and repository-pattern abstractions in NestJS to reduce query latency across high-volume collections.
- *Domain:* Inventory management, procurement workflows, POS systems, supplier reconciliation, multi-branch restaurant operations.

AI Engineer

July 2024 – June 2025

University of Texas at Arlington

Arlington, TX

- Built a ReAct agent with LangGraph that converts natural-language questions into executable SQL across a relational schema with foreign-key joins, replacing manual query writing for internal research data lookups.
- Designed a tool-calling loop with schema-aware prompting and structured output parsing, the agent self-corrects on SQL errors and retries with fixed joins, no hardcoded templates needed.
- Developed a Next.js chat interface with serialized LangChain message history and type-safe rendering of human/AI/system message types, supporting full conversation context per query turn.
- Integrated the agent into an existing research management web app (React, Flask, MySQL) that served 50 concurrent users with professor profiles and submission tracking dashboards.

ML Intern

June 2021 – Sept 2021

Exposys Data Labs

Bengaluru

- Built classification pipelines with scikit-learn (K-means, Decision Trees) and automated feature selection, increasing prediction accuracy by 24% over baseline through systematic model comparison.
- Rewrote data processing scripts using vectorized Pandas operations and memory-efficient joins, cutting ETL runtime by 30% on datasets with 50K+ records.
- Created regression and clustering dashboards in Plotly that surfaced actionable patterns for the analytics team, replacing static spreadsheet reports.

PROJECTS

Multi-PDF Conversational RAG Application — Google Gemini Pro, LangChain, FAISS, Streamlit

- Engineered a RAG pipeline with LangChain and FAISS that enables conversational Q&A across multiple PDFs, achieving sub-second retrieval on 200MB+ document sets with 90%+ grounded answers.
- Built a chunking workflow with RecursiveCharacterTextSplitter and Google embedding-001 — chose recursive splitting over fixed-size windows after testing showed 15% better retrieval relevance on long-form research papers.
- Deployed as an interactive Streamlit app with PDF upload, real-time vectorization, and a chat interface for literature review automation.

EDUCATION

University of Texas at Arlington

Texas, USA

Master of Science in Computer Science, 2022 – 2024

GPA: 3.62, Graduate Certificate in Deep Learning

- Relevant Coursework: Web Data Management, Machine Learning, Advanced Algorithms, Cloud Computing

University of Mumbai

Mumbai

Bachelor of Technology, Computer Engineering

Aug 2018 – July 2022